

FMCG ERP SYSTEM

# FMCG ERP Full Reference Manual

Complete ERP Reference — All Modules, All Workflows

Audience: All Roles — Super User Reference

Version: v2.0 — Post-Recovery Edition

Date: 24 May 2026

Status: Confidential — Internal Use Only

ERP button/link recovery complete. All navigation targets restored and verified. Smoke test: 141/141 routes passed. BVT: 0. Broken action cards: 0.

# FMCG ERP — Full Reference Manual v2

**Version:** 2.0 (post-recovery)  
**Date:** 2026-05-24  
**Audience:** All staff, Implementation Teams, Auditors, System Administrators  
**Purpose:** Complete reference guide covering all 140+ routes, every module, all permissions, API endpoints, and cross-module data flows.

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## 1. System Architecture Overview

### Technology Stack

| Layer         | Technology                                                 |
|---------------|------------------------------------------------------------|
| Frontend      | Next.js 14 (App Router) + React + TypeScript               |
| UI Components | Tailwind CSS + shadcn/ui                                   |
| Backend       | FastAPI (Python)                                           |
| Database      | PostgreSQL                                                 |
| Auth          | Cookie-based sessions ( <code>erp_access_token</code> )    |
| API Pattern   | REST; base URL: <code>http://localhost:8000/api/v1/</code> |

### Application Ports

| Service            | Port |
|--------------------|------|
| Next.js (frontend) | 3000 |
| FastAPI (backend)  | 8000 |
| PostgreSQL         | 5432 |

### Frontend Architecture

- **App Router:** All pages in `frontend/src/app/dashboard/`
- **Middleware:** `frontend/src/middleware.ts` — handles auth guard and redirects
- **API Client:** `frontend/src/lib/api.ts` — Axios instance with base URL from `NEXT_PUBLIC_API_URL`

- Auth cookie: `erp_access_token` — required for all protected routes

## 2. Complete Route Reference

All 141 application routes (excluding login):

### Admin Routes

| Route                                       | Page            | Description                  |
|---------------------------------------------|-----------------|------------------------------|
| <code>/dashboard/admin</code>               | Admin Hub       | Administration dashboard     |
| <code>/dashboard/admin/users</code>         | Users           | User account management      |
| <code>/dashboard/admin/roles</code>         | Roles           | Role configuration           |
| <code>/dashboard/admin/permissions</code>   | Permissions     | Permission matrix            |
| <code>/dashboard/admin/security</code>      | Security        | Security policy settings     |
| <code>/dashboard/admin/system-config</code> | System Config   | Core system parameters       |
| <code>/dashboard/admin/companies</code>     | Companies       | Multi-company setup          |
| <code>/dashboard/admin/approvals</code>     | Approval Config | Approval chain configuration |
| <code>/dashboard/admin/logs</code>          | Admin Logs      | Full audit log               |

### Products & Materials

| Route                             | Page      | Description                      |
|-----------------------------------|-----------|----------------------------------|
| <code>/dashboard/products</code>  | Products  | Finished goods and trading items |
| <code>/dashboard/materials</code> | Materials | Raw materials and packaging      |

## Supply Chain Routes

| Route                                                 | Page             | Description               |
|-------------------------------------------------------|------------------|---------------------------|
| <a href="#">/dashboard/procurement</a>                | Procurement Hub  | All procurement functions |
| <a href="#">/dashboard/procurement/requests</a>       | PRs              | Purchase requisitions     |
| <a href="#">/dashboard/procurement/orders</a>         | POs              | Purchase orders           |
| <a href="#">/dashboard/procurement/rfq</a>            | RFQ              | Request for quotation     |
| <a href="#">/dashboard/procurement/deliveries</a>     | Deliveries       | Goods receipt             |
| <a href="#">/dashboard/procurement/suppliers</a>      | Suppliers        | Supplier master           |
| <a href="#">/dashboard/procurement/blanket</a>        | Blanket          | Framework agreements      |
| <a href="#">/dashboard/procurement/reorder</a>        | Reorder          | Reorder policies          |
| <a href="#">/dashboard/procurement/suggestions</a>    | Suggestions      | Auto-reorder suggestions  |
| <a href="#">/dashboard/procurement/subcontracting</a> | Subcontracting   | Outsourced orders         |
| <a href="#">/dashboard/procurement/landed-cost</a>    | Landed Cost      | Import cost allocation    |
| <a href="#">/dashboard/suppliers</a>                  | Suppliers Master | Full supplier management  |
| <a href="#">/dashboard/warehouses</a>                 | Warehouses       | Warehouse configuration   |
| <a href="#">/dashboard/warehouses/wms</a>             | WMS              | Warehouse management      |
| <a href="#">/dashboard/inventory</a>                  | Inventory        | Stock management          |
| <a href="#">/dashboard/inventory/stock</a>            | Stock            | Current stock levels      |
| <a href="#">/dashboard/inventory/movements</a>        | Movements        | Transaction history       |
| <a href="#">/dashboard/inventory/cycle-count</a>      | Cycle Count      | Physical stock count      |
| <a href="#">/dashboard/inventory/shelf-life</a>       | Shelf Life       | Expiry tracking           |
| <a href="#">/dashboard/inventory/traceability</a>     | Traceability     | Batch/lot trace           |
| <a href="#">/dashboard/inventory/serials</a>          | Serials          | Serial number tracking    |
| <a href="#">/dashboard/inventory/valuation</a>        | Valuation        | Stock valuation           |
| <a href="#">/dashboard/wms</a>                        | WMS Hub          | WMS main                  |
| <a href="#">/dashboard/wms/zones</a>                  | Zones            | Warehouse zones           |
| <a href="#">/dashboard/wms/locations</a>              | Locations        | Bin locations             |
| <a href="#">/dashboard/wms/quarantine</a>             | Quarantine       | Quarantine hold           |
| <a href="#">/dashboard/logistics</a>                  | Logistics        | Shipment tracking         |

## Manufacturing Routes

| Route                                                                                               | Page           | Description                 |
|-----------------------------------------------------------------------------------------------------|----------------|-----------------------------|
| <a href="/dashboard/recipes">/dashboard/recipes</a>                                                 | Recipes        | Product formulations        |
| <a href="/dashboard/recipes/{id}">/dashboard/recipes/{id}</a>                                       | Recipe Detail  | Recipe detail and BOM items |
| <a href="/dashboard/bom">/dashboard/bom</a>                                                         | BOM            | Bill of materials           |
| <a href="/dashboard/production">/dashboard/production</a>                                           | Production Hub | Production workspace        |
| <a href="/dashboard/production/orders">/dashboard/production/orders</a>                             | Work Orders    | Production orders           |
| <a href="/dashboard/production/execution">/dashboard/production/execution</a>                       | Execution      | Real-time execution         |
| <a href="/dashboard/production/material-flow">/dashboard/production/material-flow</a>               | Material Flow  | Material consumption        |
| <a href="/dashboard/production/costing">/dashboard/production/costing</a>                           | Costing        | Actual cost tracking        |
| <a href="/dashboard/shop-floor">/dashboard/shop-floor</a>                                           | Shop Floor     | Shop floor management       |
| <a href="/dashboard/planning">/dashboard/planning</a>                                               | Planning       | MRP/MPS/capacity            |
| <a href="/dashboard/planning/mrp">/dashboard/planning/mrp</a>                                       | MRP            | Material requirements       |
| <a href="/dashboard/planning/mps">/dashboard/planning/mps</a>                                       | MPS            | Master production schedule  |
| <a href="/dashboard/npd">/dashboard/npd</a>                                                         | NPD            | New product development     |
| <a href="/dashboard/quality">/dashboard/quality</a>                                                 | Quality        | QC management               |
| <a href="/dashboard/quality/inspections">/dashboard/quality/inspections</a>                         | Inspections    | QC inspections              |
| <a href="/dashboard/quality/certificates">/dashboard/quality/certificates</a>                       | Certificates   | Quality certificates        |
| <a href="/dashboard/quality/parameters">/dashboard/quality/parameters</a>                           | Parameters     | Test parameters             |
| <a href="/dashboard/compliance">/dashboard/compliance</a>                                           | Compliance     | Regulatory compliance       |
| <a href="/dashboard/compliance?tab=gs1">/dashboard/compliance?tab=gs1</a>                           | GS1 Labels     | Barcode generation          |
| <a href="/dashboard/compliance?tab=regulatory-certs">/dashboard/compliance?tab=regulatory-certs</a> | Reg Certs      | Regulatory certificates     |

## Sales Routes

| Route                                                                                   | Page         | Description            |
|-----------------------------------------------------------------------------------------|--------------|------------------------|
| <a href="/dashboard/sales">/dashboard/sales</a>                                         | Sales Hub    | Full sales workspace   |
| <a href="/dashboard/sales/orders">/dashboard/sales/orders</a>                           | Orders       | Sales orders           |
| <a href="/dashboard/sales/invoices">/dashboard/sales/invoices</a>                       | Invoices     | Customer invoices      |
| <a href="/dashboard/sales/customers">/dashboard/sales/customers</a>                     | Customers    | Customer master        |
| <a href="/dashboard/sales/quotes">/dashboard/sales/quotes</a>                           | Quotes       | Quotations             |
| <a href="/dashboard/sales/pricing">/dashboard/sales/pricing</a>                         | Pricing      | Price management       |
| <a href="/dashboard/sales/contracts">/dashboard/sales/contracts</a>                     | Contracts    | Sales agreements       |
| <a href="/dashboard/sales/recurring-orders">/dashboard/sales/recurring-orders</a>       | Recurring    | Auto-scheduled orders  |
| <a href="/dashboard/sales/van-sales">/dashboard/sales/van-sales</a>                     | Van Sales    | Van sales management   |
| <a href="/dashboard/sales/van-sales/vans/{id}">/dashboard/sales/van-sales/vans/{id}</a> | Van Detail   | Individual van         |
| <a href="/dashboard/sales/distributors">/dashboard/sales/distributors</a>               | Distributors | Distributor management |
| <a href="/dashboard/sales/field-sales">/dashboard/sales/field-sales</a>                 | Field Sales  | Field rep management   |
| <a href="/dashboard/pos">/dashboard/pos</a>                                             | POS          | Point of sale          |

## Finance Routes

| Route                                                                       | Page          | Description            |
|-----------------------------------------------------------------------------|---------------|------------------------|
| <a href="/dashboard/finance">/dashboard/finance</a>                         | Finance Hub   | Full finance workspace |
| <a href="/dashboard/cashbook">/dashboard/cashbook</a>                       | Cashbook      | Cash and bank accounts |
| <a href="/dashboard/payroll">/dashboard/payroll</a>                         | Payroll       | Payroll management     |
| <a href="/dashboard/expenses">/dashboard/expenses</a>                       | Expenses      | Expense claims         |
| <a href="/dashboard/fixed-assets">/dashboard/fixed-assets</a>               | Fixed Assets  | Asset register         |
| <a href="/dashboard/bank-reconciliation">/dashboard/bank-reconciliation</a> | Bank Recon    | Bank reconciliation    |
| <a href="/dashboard/bank-api">/dashboard/bank-api</a>                       | Bank API      | Bank feed integration  |
| <a href="/dashboard/tax">/dashboard/tax</a>                                 | Tax           | Tax management         |
| <a href="/dashboard/contracts">/dashboard/contracts</a>                     | Contracts     | Financial contracts    |
| <a href="/dashboard/invoice-match">/dashboard/invoice-match</a>             | Invoice Match | 3-way match            |
| <a href="/dashboard/etims">/dashboard/etims</a>                             | eTIMS         | KRA tax submissions    |
| <a href="/dashboard/integrations">/dashboard/integrations</a>               | Integrations  | External systems       |
| <a href="/dashboard/integrations/mpesa">/dashboard/integrations/mpesa</a>   | M-Pesa        | M-Pesa configuration   |

## HR Routes

| Route                                                         | Page      | Description       |
|---------------------------------------------------------------|-----------|-------------------|
| <a href="/dashboard/hr">/dashboard/hr</a>                     | HR Hub    | Full HR workspace |
| <a href="/dashboard/hr/employees">/dashboard/hr/employees</a> | Employees | Employee master   |

## CRM & Marketing Routes

| Route                                | Page      | Description                      |
|--------------------------------------|-----------|----------------------------------|
| <a href="#">/dashboard/crm</a>       | CRM       | Customer relationship management |
| <a href="#">/dashboard/marketing</a> | Marketing | Campaign and TPM management      |

## Maintenance & Utilities Routes

| Route                                             | Page        | Description           |
|---------------------------------------------------|-------------|-----------------------|
| <a href="#">/dashboard/maintenance</a>            | Maintenance | Equipment maintenance |
| <a href="#">/dashboard/maintenance/assets</a>     | Assets      | Asset register        |
| <a href="#">/dashboard/maintenance/breakdowns</a> | Breakdowns  | Breakdown log         |
| <a href="#">/dashboard/utility-management</a>     | Utilities   | Electricity and water |

## Analytics & AI Routes

| Route                                               | Page                 | Description           |
|-----------------------------------------------------|----------------------|-----------------------|
| <a href="#">/dashboard/analytics</a>                | Analytics            | Business intelligence |
| <a href="#">/dashboard/analytics/sales</a>          | Sales Analytics      | Sales analysis        |
| <a href="#">/dashboard/analytics/production</a>     | Production Analytics | Manufacturing KPIs    |
| <a href="#">/dashboard/analytics/inventory</a>      | Inventory Analytics  | Stock analysis        |
| <a href="#">/dashboard/analytics/finance</a>        | Finance Analytics    | Financial reporting   |
| <a href="#">/dashboard/analytics/report-builder</a> | Report Builder       | Custom reports        |
| <a href="#">/dashboard/ai</a>                       | AI                   | AI assistant          |

## Documents & Communication Routes

| Route                                               | Page           | Description        |
|-----------------------------------------------------|----------------|--------------------|
| <a href="#">/dashboard/documents</a>                | Documents      | Document library   |
| <a href="#">/dashboard/documents/knowledge-base</a> | Knowledge Base | SOPs and articles  |
| <a href="#">/dashboard/communication</a>            | Communication  | Internal messaging |
| <a href="#">/dashboard/helpdesk</a>                 | Helpdesk       | Support tickets    |
| <a href="#">/dashboard/approvals</a>                | Approvals      | Approval queue     |
| <a href="#">/dashboard/logs</a>                     | Logs           | System audit log   |
| <a href="#">/dashboard/logs/security</a>            | Security Logs  | Security events    |

## 3. Module Reference Index

Quick-reference table of all modules with their manual, route, and primary audience:

| Module        | Manual               | Primary Route                                                             | Primary User         |
|---------------|----------------------|---------------------------------------------------------------------------|----------------------|
| Manufacturing | Manufacturing Manual | <a href="/dashboard/production">/dashboard/production</a>                 | Production Manager   |
| Recipes       | Manufacturing Manual | <a href="/dashboard/recipes">/dashboard/recipes</a>                       | Production / NPD     |
| BOM           | Manufacturing Manual | <a href="/dashboard/bom">/dashboard/bom</a>                               | Production Manager   |
| Planning      | Manufacturing Manual | <a href="/dashboard/planning">/dashboard/planning</a>                     | Planning Manager     |
| Shop Floor    | Manufacturing Manual | <a href="/dashboard/shop-floor">/dashboard/shop-floor</a>                 | Supervisor           |
| NPD           | Manufacturing Manual | <a href="/dashboard/npd">/dashboard/npd</a>                               | NPD Manager          |
| Quality       | Manufacturing Manual | <a href="/dashboard/quality">/dashboard/quality</a>                       | Quality Officer      |
| Compliance    | Manufacturing Manual | <a href="/dashboard/compliance">/dashboard/compliance</a>                 | Quality / Regulatory |
| Procurement   | Supply Chain Manual  | <a href="/dashboard/procurement">/dashboard/procurement</a>               | Procurement Officer  |
| Inventory     | Supply Chain Manual  | <a href="/dashboard/inventory">/dashboard/inventory</a>                   | Warehouse Manager    |
| WMS           | Supply Chain Manual  | <a href="/dashboard/wms">/dashboard/wms</a>                               | Warehouse Manager    |
| Logistics     | Supply Chain Manual  | <a href="/dashboard/logistics">/dashboard/logistics</a>                   | Logistics Officer    |
| Sales         | Sales Manual         | <a href="/dashboard/sales">/dashboard/sales</a>                           | Sales Manager        |
| POS           | Sales Manual         | <a href="/dashboard/pos">/dashboard/pos</a>                               | Cashier              |
| Finance       | Finance Manual       | <a href="/dashboard/finance">/dashboard/finance</a>                       | Accountant           |
| Payroll       | Finance Manual       | <a href="/dashboard/payroll">/dashboard/payroll</a>                       | Payroll Officer      |
| HR            | HR Manual            | <a href="/dashboard/hr">/dashboard/hr</a>                                 | HR Officer           |
| CRM           | Commercial Manual    | <a href="/dashboard/crm">/dashboard/crm</a>                               | Sales / CRM          |
| Marketing     | Commercial Manual    | <a href="/dashboard/marketing">/dashboard/marketing</a>                   | Marketing Manager    |
| Maintenance   | Maintenance Manual   | <a href="/dashboard/maintenance">/dashboard/maintenance</a>               | Engineer             |
| Utilities     | Maintenance Manual   | <a href="/dashboard/utility-management">/dashboard/utility-management</a> | Utility Manager      |
| Analytics     | Intelligence Manual  | <a href="/dashboard/analytics">/dashboard/analytics</a>                   | All Managers         |
| AI            | Intelligence Manual  | <a href="/dashboard/ai">/dashboard/ai</a>                                 | All Staff            |
| Admin         | Admin Manual         | <a href="/dashboard/admin">/dashboard/admin</a>                           | System Admin         |
| Documents     | Documents Manual     | <a href="/dashboard/documents">/dashboard/documents</a>                   | All Staff            |
| Helpdesk      | Documents Manual     | <a href="/dashboard/helpdesk">/dashboard/helpdesk</a>                     | All Staff            |

## 4. Permission Reference

### Permission Naming Convention

`<module>.<resource>.<action>`

Examples: `procurement.purchase_orders.approve` , `inventory.stock.adjust` , `hr.payroll.view`



## Actions Available Per Resource

| Action  | Meaning                                  |
|---------|------------------------------------------|
| view    | Read access — see lists and details      |
| create  | Create new records                       |
| edit    | Update existing records                  |
| delete  | Delete records (usually DRAFT only)      |
| approve | Approve workflows (journals, POs, leave) |
| export  | Export to Excel/CSV/PDF                  |
| admin   | Full access including configuration      |

## Critical Permissions

| Permission                          | Risk Level | Who Should Have         |
|-------------------------------------|------------|-------------------------|
| <code>finance.journals.post</code>  | High       | Finance Manager only    |
| <code>finance.invoices.void</code>  | High       | Finance Manager only    |
| <code>sales.invoices.void</code>    | High       | Sales Manager + Finance |
| <code>inventory.stock.adjust</code> | Medium     | Warehouse Manager       |
| <code>hr.payroll.approve</code>     | High       | Finance Manager / MD    |
| <code>admin.users.create</code>     | High       | System Admin only       |
| <code>admin.permissions.edit</code> | Critical   | System Admin only       |
| <code>logs.export</code>            | Medium     | Finance Manager + Admin |

## 5. Data Flow Reference

### Order-to-Cash Flow

```

CRM Lead → Opportunity →
Sales Quote → Sales Order (Confirmed) →
WMS Pick Wave → Dispatch (Inventory decrease) →
Delivery Confirmation →
Sales Invoice (Posted) →
eTIMS Submission (KRA) →
Customer Payment (Collection) →
Bank Reconciliation →
AR Cleared

```

### Procure-to-Pay Flow

```

MRP/Suggestion → Purchase Requisition (Draft) →
PR Approval →
Purchase Order (Sent to Supplier) →
Goods Receipt (GRN) → Inventory Increase →
Invoice Match (3-way: PO + GRN + Supplier Invoice) →
AP Invoice Posted →
Supplier Payment →
Bank Reconciliation →
AP Cleared

```

## Production Flow

```
MPS/Demand → Production Plan →  
Work Order (Released) →  
Shop Floor Execution →  
Material Issue (Inventory Decrease) →  
Work Order Completion →  
Finished Goods Inventory Increase →  
QC Inspection →  
PASS: Available for Sale | FAIL: Quarantine
```

## Payroll Flow

```
Employee Master Data →  
Attendance Records →  
Leave Records →  
Payroll Profile (Salary + Deductions) →  
Payroll Run (PAYE / NSSF / NHIF computed) →  
Payroll Approval →  
GL Journal Post (Salaries Payable) →  
Bank Payment File Generated →  
Employee Payment →  
Payslip Available on ESS
```

---

## 6. API Endpoints Reference

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Base URL: <http://localhost:8000/api/v1/>

## Core Endpoint Patterns

| Resource         | List                      | Create                     | Get                           | Update                          | Delete                            |
|------------------|---------------------------|----------------------------|-------------------------------|---------------------------------|-----------------------------------|
| Products         | GET /products/            | POST /products/            | GET /products/{id}            | PATCH /products/{id}            | DELETE /products/{id}             |
| Materials        | GET /materials/           | POST /materials/           | GET /materials/{id}           | PATCH /materials/{id}           | DELETE /materials/{id}            |
| Suppliers        | GET /suppliers/           | POST /suppliers/           | GET /suppliers/{id}           | PATCH /suppliers/{id}           | —                                 |
| Purchase Orders  | GET /purchase-orders/     | POST /purchase-orders/     | GET /purchase-orders/{id}     | PATCH /purchase-orders/{id}     | —                                 |
| Sales Orders     | GET /sales-orders/        | POST /sales-orders/        | GET /sales-orders/{id}        | PATCH /sales-orders/{id}        | —                                 |
| Recipes          | GET /recipes/             | POST /recipes/             | GET /recipes/{id}             | PATCH /recipes/{id}             | DELETE /recipes/{id} (DRAFT only) |
| Work Orders      | GET /work-orders/         | POST /work-orders/         | GET /work-orders/{id}         | PATCH /work-orders/{id}         | —                                 |
| Employees        | GET /employees/           | POST /employees/           | GET /employees/{id}           | PATCH /employees/{id}           | —                                 |
| Inventory Stock  | GET /inventory/stock/     | POST /inventory/stock/     | GET /inventory/stock/{id}     | PATCH /inventory/stock/{id}     | —                                 |
| Invoices         | GET /invoices/            | POST /invoices/            | GET /invoices/{id}            | PATCH /invoices/{id}            | —                                 |
| QC Inspections   | GET /quality/inspections/ | POST /quality/inspections/ | GET /quality/inspections/{id} | PATCH /quality/inspections/{id} | —                                 |
| Regulatory Certs | GET /regulatory-certs/    | POST /regulatory-certs/    | GET /regulatory-certs/{id}    | PATCH /regulatory-certs/{id}    | DELETE /regulatory-certs/{id}     |

## Authentication

All API calls require authentication. The frontend uses cookie-based auth ( `erp_access_token` ). For direct API calls, include the cookie in the request header.

**Login:** `POST /api/v1/auth/login` with `{email, password}` → returns cookie

## API Response Formats

**Success (200):** `{data: {...}, status: "success"}`

**List (200):** `{data: [...], total: N, page: N, per_page: N}`

**Create (201):** `{data: {...}, status: "created"}`

**Error (4xx/5xx):** `{detail: "error message", status: "error"}`

## 7. Status Code Reference

### Universal Status Patterns

| Status              | HTTP | Meaning                          |
|---------------------|------|----------------------------------|
| DRAFT               | —    | Record created but not submitted |
| SUBMITTED / PENDING | —    | Awaiting approval                |
| APPROVED            | —    | Approved and ready               |
| ACTIVE              | —    | In use / live                    |
| COMPLETED           | —    | Process finished                 |
| CLOSED              | —    | Fully reconciled and locked      |
| CANCELLED           | —    | Cancelled; no further action     |
| VOID                | —    | Voided; reversed                 |

### Document-Specific Statuses

**Purchase Orders:** DRAFT → SENT → ACKNOWLEDGED → PARTIAL\_DELIVERY → DELIVERED → INVOICED → CLOSED

**Sales Orders:** DRAFT → CONFIRMED → PICKING → DISPATCHED → DELIVERED → INVOICED → CLOSED

**Work Orders:** DRAFT → CONFIRMED → RELEASED → IN\_PROGRESS → COMPLETED → CLOSED

**Recipes:** DRAFT → APPROVED → OBSOLETE

**Invoices:** DRAFT → POSTED → PARTIALLY\_PAID → PAID → VOID

**Leave:** DRAFT → SUBMITTED → APPROVED → REJECTED

**Payroll:** DRAFT → GENERATED → APPROVED → POSTED → PAID

## 8. Field Reference — Common Fields

Fields used across many modules:

| Field                   | Type     | Pattern                                     |
|-------------------------|----------|---------------------------------------------|
| <code>id</code>         | UUID     | Unique identifier — never reused            |
| <code>created_at</code> | datetime | ISO 8601, UTC                               |
| <code>updated_at</code> | datetime | ISO 8601, UTC                               |
| <code>is_active</code>  | bool     | Default true; false = soft-deleted/inactive |
| <code>status</code>     | str      | Module-specific status string               |
| <code>reference</code>  | str      | Human-readable reference number             |
| <code>notes</code>      | str      | Free text annotation                        |
| <code>company_id</code> | UUID     | Multi-company isolation                     |

**Soft Delete Pattern:** Most records use `is_active = false` rather than hard delete. This preserves audit trail.

**Currency Fields:** Stored as Decimal(15,2). Always in the document's currency. Multi-currency: separate `amount_base` field stores KES equivalent.

## 9. Keyboard Shortcuts & UI Tips

### Navigation

- **Ctrl+K** or **/**: Open global search
- **Esc**: Close modal or dialog
- **Tab**: Move between form fields
- **Enter**: Submit focused button

### Tips

- **Filter persistence**: Date range filters persist within a session — check them when results look wrong
- **Tab memory**: The ERP remembers your last open tab per module within a session
- **Notifications**: Bell icon top-right shows count of pending approvals and alerts
- **Export all lists**: Every list has an Export button → downloads current filtered view as Excel
- **Print PDFs**: Open document → Print → select "Save as PDF" in browser print dialog

## 10. Configuration Reference

### Environment Variables (Frontend)

| Variable                          | Purpose              | Default                            |
|-----------------------------------|----------------------|------------------------------------|
| <code>NEXT_PUBLIC_API_URL</code>  | Backend API base URL | <code>http://localhost:8000</code> |
| <code>NEXT_PUBLIC_APP_NAME</code> | Application title    | <code>FMCG ERP</code>              |

### Environment Variables (Backend)

| Variable                           | Purpose                             |
|------------------------------------|-------------------------------------|
| <code>DATABASE_URL</code>          | PostgreSQL connection string        |
| <code>SECRET_KEY</code>            | JWT signing key                     |
| <code>API_V1_STR</code>            | API prefix ( <code>/api/v1</code> ) |
| <code>MPESA_CONSUMER_KEY</code>    | Safaricom Daraja key                |
| <code>MPESA_CONSUMER_SECRET</code> | Safaricom Daraja secret             |
| <code>MPESA_SHORTCODE</code>       | Till/Paybill number                 |
| <code>ETIMS_API_KEY</code>         | KRA eTIMS credential                |

### Middleware Configuration

`frontend/src/middleware.ts` — handles:

1. Auth guard: Redirect to `/login` if no `erp_access_token` cookie
2. Dashboard consolidation: Redirect legacy routes to canonical paths
3. All redirects use HTTP 302 with `Cache-Control: no-store` headers

## 11. Glossary

| Term               | Definition                                                                                 |
|--------------------|--------------------------------------------------------------------------------------------|
| <b>3-Way Match</b> | Verification that Purchase Order, Goods Receipt, and Supplier Invoice agree before payment |
| <b>BOM</b>         | Bill of Materials — list of components and quantities to produce a product                 |
| <b>COA</b>         | Chart of Accounts — numbered list of all GL accounts                                       |
| <b>DSO</b>         | Days Sales Outstanding — average days to collect payment after invoicing                   |
| <b>eTIMS</b>       | Electronic Tax Invoice Management System — KRA's electronic invoicing system               |
| <b>GRN</b>         | Goods Receipt Note — document confirming goods received from supplier                      |
| <b>MTBF</b>        | Mean Time Between Failures — average time equipment runs between breakdowns                |
| <b>MTTR</b>        | Mean Time To Repair — average time to restore equipment after failure                      |
| <b>MPS</b>         | Master Production Schedule — planned production quantities by period                       |
| <b>MRP</b>         | Material Requirements Planning — calculation of what to purchase based on demand           |
| <b>NCMR</b>        | Non-Conformance Material Report — record of product failing QC                             |
| <b>OEE</b>         | Overall Equipment Effectiveness — Availability × Performance × Quality                     |
| <b>OTIF</b>        | On Time In Full — % of deliveries arriving on scheduled date with full quantity            |
| <b>PAYE</b>        | Pay As You Earn — employee income tax deducted at source                                   |
| <b>PO</b>          | Purchase Order — formal order sent to a supplier                                           |
| <b>PR / PRQ</b>    | Purchase Requisition — internal request to purchase something                              |
| <b>RFQ</b>         | Request for Quotation — soliciting price quotes from suppliers                             |
| <b>SKU</b>         | Stock Keeping Unit — unique product identifier                                             |
| <b>SLA</b>         | Service Level Agreement — committed response/resolution time                               |
| <b>SOH</b>         | Stock on Hand — physical inventory quantity currently in warehouse                         |
| <b>TPM</b>         | Trade Promotion Management — management of promotional spend in retail channels            |
| <b>UOM</b>         | Unit of Measure — e.g. KG, Litre, Each, Carton                                             |
| <b>VAT</b>         | Value Added Tax — 16% standard rate in Kenya                                               |
| <b>WIP</b>         | Work in Process — materials committed to a work order but not yet completed                |
| <b>WMS</b>         | Warehouse Management System — software for managing warehouse operations                   |

*End of Full Reference Manual v2*

## Quick Module Cross-Reference

For detailed guides on each module, refer to the respective manual:

| Manual                         | File                                                                                      |
|--------------------------------|-------------------------------------------------------------------------------------------|
| Manufacturing                  | <a href="#">docs/manuals/manufacturing/FMCG-ERP-Manufacturing-Manual-v2.md</a>            |
| Supply Chain                   | <a href="#">docs/manuals/supply-chain/FMCG-ERP-Supply-Chain-Manual-v2.md</a>              |
| Sales & Distribution           | <a href="#">docs/manuals/sales-distribution/FMCG-ERP-Sales-Distribution-Manual-v2.md</a>  |
| Commercial (CRM/Marketing)     | <a href="#">docs/manuals/commercial/FMCG-ERP-Commercial-CRM-Marketing-Manual-v2.md</a>    |
| Finance & Payroll              | <a href="#">docs/manuals/finance-payroll/FMCG-ERP-Finance-Payroll-Manual-v2.md</a>        |
| HR                             | <a href="#">docs/manuals/hr/FMCG-ERP-HR-Manual-v2.md</a>                                  |
| Logistics                      | <a href="#">docs/manuals/logistics/FMCG-ERP-Logistics-Manual-v2.md</a>                    |
| Maintenance & Utilities        | <a href="#">docs/manuals/maintenance/FMCG-ERP-Maintenance-Utilities-Manual-v2.md</a>      |
| Documents & Communication      | <a href="#">docs/manuals/documents/FMCG-ERP-Documents-Communication-Manual-v2.md</a>      |
| Administration                 | <a href="#">docs/manuals/admin/FMCG-ERP-Administration-Manual-v2.md</a>                   |
| Intelligence & Analytics       | <a href="#">docs/manuals/intelligence/FMCG-ERP-Intelligence-Analytics-AI-Manual-v2.md</a> |
| Kenya Go-Live Training         | <a href="#">docs/manuals/kenya-go-live/Kenya-Go-Live-ERP-Training-Manual-v2.md</a>        |
| Full Reference (this document) | <a href="#">docs/manuals/full-reference/FMCG-ERP-Full-Reference-Manual-v2.md</a>          |